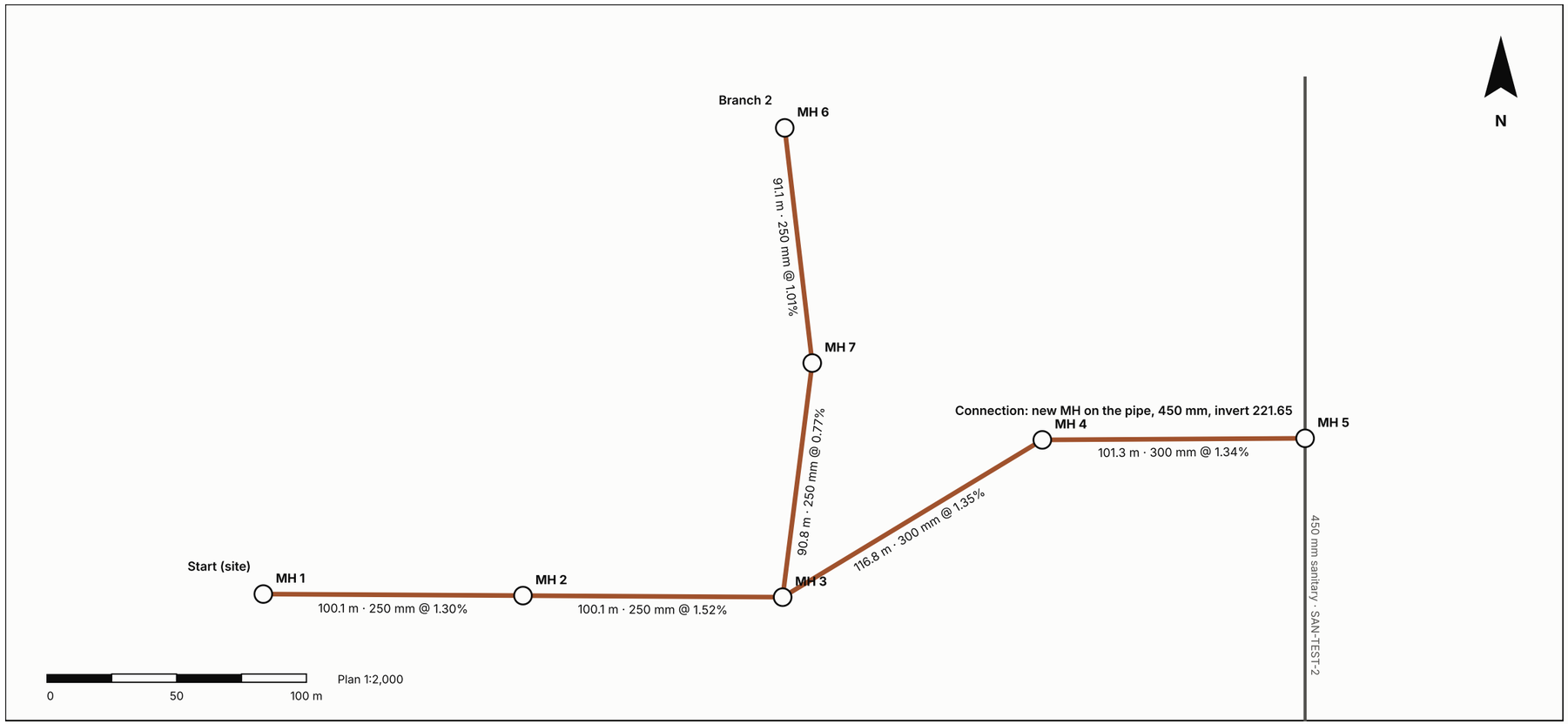
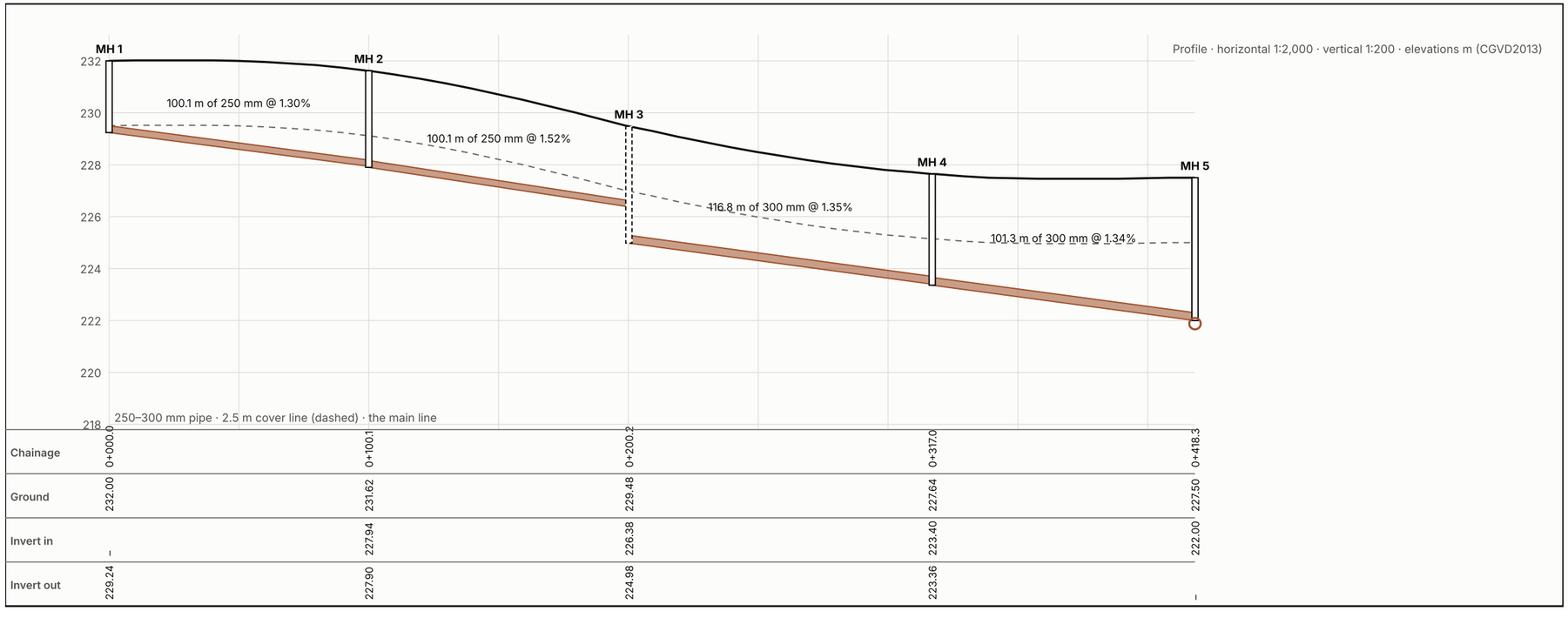




Chainage	0+000.0	0+100.1	0+200.2	0+317.0	0+418.3
Ground	232.00	231.62	229.48	227.64	227.50
Invert in		227.94	226.38	223.40	222.00
Invert out	229.24	227.90	224.98	223.36	

DESIGN

A **250–300 mm** sewer (the main line and 1 branch) at 0.77–1.52%, 7 maintenance holes, 2.8 m deep at the start and 5.5 m at the deepest, for 78.3 L/s (60.1 L/s peak dry + 18.2 L/s I&I): the least excavation the rules allow.

LEGEND

- The new sewer (250–300 mm)
- The existing sanitary sewer it connects to
- Other Region sanitary sewers
- Maintenance hole (numbered from the start)
- Ground (profile)
- 2.5 m cover line (profile)
- Drop structure (dashed maintenance hole)

RULES CHECKED

- ✓ Pipe size (200 mm minimum, no larger than the existing sewer, never a larger pipe into a smaller one): 250–300 mm, into 450 mm
- ✓ Velocity at the design flow (0.75 m/s or more): 1.15 m/s at the slowest
- ✓ Velocity flowing full (3.0 m/s or less): 1.59 m/s at the fastest
- ✓ Flow ratio, design flow to full flow (0.7 or less): 0.70
- ✓ The last leg, at the upstream end of each branch (1.00% or more): 1.01% at the least
- ✓ Cover over the pipe (2.5 m or more): 2.51 m at the least
- ✓ Trench (8.0 m or less): 5.65 m at the deepest
- ✓ Maintenance holes (200 m apart or less): 117 m at the longest
- ✓ Drops through the maintenance holes (0.03–0.20 m, obverts matched where the size grows, or a drop structure from 0.9 m): 1 drop structure
- ✓ Connection (the new obvert at or above the existing one): 0.20 m drop

6 reaches with supercritical flow (Froude 1 or more): the standard asks for subcritical flow where possible (s. 2.9.1).

PRELIMINARY · NOT FOR CONSTRUCTION

Preliminary sanitary sewer Plan and profile

Location	From 43.66999, -79.84262
Connection	450 mm sanitary sewer SAN-TEST-2, invert 221.65 m (new maintenance hole on the pipe)
Design flow	78.3 L/s (60.1 L/s peak dry + 18.2 L/s I&I)
Sewer	250–300 mm at 0.77–1.52%, the main line and 1 branch, 7 maintenance holes, 600 m; 2.8 m deep at the start, 5.5 m at the deepest; 1 drop structure
Scales	Plan 1:2,000 · profile 1:2,000 H, 1:200 V, on 11×17 at 100%
Ground	Ontario lidar terrain model (Test lidar, 0.5 m) (not a survey)
Sewers	Region of Peel sanitary mains (open data), read October 17, 2026
Rules	Region of Peel Linear Wastewater Standards (published Nov 2022, R1.0 Mar 2023), s. 3.1–3.4 and 6.1

General screening results only. Peel's W&WW Infrastructure Planning team must validate or confirm all results.

October 17, 2026 **Sheet 1 of 2**

Unofficial prototype · public data only · not a Region of Peel product · Labs · vdev

MAINTENANCE HOLES

MH	Chainage	Ground (m)	Invert in (m)	Invert out (m)	Depth to invert (m)	Drop (m)	
MH 1 (start)	0+000.0	232.00	–	229.24	2.76	–	
MH 2	0+100.1	231.62	227.94	227.90	3.72	0.04	
MH 3 (junction)	0+200.2	229.48	226.38 / 225.04	224.98	4.50	1.40	drop structure
MH 4	0+317.0	227.64	223.40	223.36	4.28	0.04	
MH 5 (connection)	0+418.3	227.50	222.00	–	5.50	0.20	
MH 6 (branch 2 start)	branch 2 0+000.0	229.47	–	226.70	2.77	–	
MH 7	branch 2 0+091.1	229.20	225.78	225.74	3.46	0.04	

REACHES

Reach	Diameter (mm)	Flow (L/s)	Length (m)	Slope (%)	Invert up (m)	Invert down (m)	Full flow (L/s)	Flow ratio	V design (m/s)	V full (m/s)	Depth ratio	Froude
MH 1 → MH 2	250	47.1	100.1	1.30	229.24	227.94	67.8	0.70	1.49	1.38	0.61	1.32
MH 2 → MH 3	250	47.1	100.1	1.52	227.90	226.38	73.3	0.64	1.59	1.49	0.58	1.46
MH 3 → MH 4	300	78.3	116.8	1.35	224.98	223.40	112.5	0.70	1.72	1.59	0.61	1.39
MH 4 → MH 5	300	78.3	101.3	1.34	223.36	222.00	112.1	0.70	1.71	1.59	0.62	1.38
MH 6 → MH 7	250	36.2	91.1	1.01	226.70	225.78	59.8	0.61	1.27	1.22	0.56	1.20
MH 7 → MH 3	250	36.2	90.8	0.77	225.74	225.04	52.2	0.69	1.15	1.06	0.61	1.02

NOTES

- A preliminary design for screening: the route as drawn, the least excavation the rules allow, each reach the smallest size that carries its flow (never smaller than a pipe flowing into it). Not a detailed design and not for construction.
- Ground from Ontario lidar terrain model (Test lidar, 0.5 m), not a survey; cover is measured from the ground along the line, not the road centreline (s. 3.4.1.1). The existing sewer's invert and size are as the Region publishes them.
- Design: Manning's n 0.013; 0.75 m/s or more at the design flow; 3.0 m/s or less flowing full; design flow 0.7 of full flow or less; the last leg 1.00% or more; 2.5 m of cover over the pipe; trench 8.0 m or less (0.15 m bedding); maintenance holes 200 m apart or less (190 m from 600 mm); drops 0.03–0.20 m, or a drop structure from 0.9 m; the new obvert at or above the existing one.
- Not considered: utilities and crossings, soils, groundwater, easements, construction method, lands external to the site (s. 3.4.1.1).
- Drawn without a base map. Print at 100% (no "fit to page") on 11×17 landscape to keep the scales; the scale bar holds at any size.

General screening results only. Peel's W&WW Infrastructure Planning team must validate or confirm all results.